Yi Qiu

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EDUCATION

The Pennsylvania State University, *Pennsylvania, United States* — *2022 – 2027 (expected)*
Ph.D. in Physics — Advisors: David Radice

Nanjing University, *Nanjing, China* — *2021 – 2022*
Research Assistant in Astronomy — Advisors: Jian-Hua He

Dalian University of Technology, *Dalian, China* — *2017 – 2021*
Bachelor of Science in Applied Physics — Advisors: Weijie Fu, Lixin Xu

PUBLICATIONS

1. **Yi Qiu***, David Radice, Raffaele Ferrari, Albino Perego, *"A Hybrid Neural Network Emulator for M1 Neutrino Reaction Rates in Binary Neutron Star Merger Simulations."* arXiv:2610.28740
2. **Yi Qiu***, David Radice, Sherwood Richers, Federico Maria Guercilena, Albino Perego, Maitraya Bhattacharyya, *"Impact of Neutrino Flavor Conversions on Neutron Star Merger Dynamics, Ejecta, Nucleosynthesis, and Multi-Messenger Signals."* Phys.Rev.D 112 (2025) 12, 123039, arXiv:2510.15028
3. **Yi Qiu***, David Radice, Sherwood Richers, Maitraya Bhattacharyya, *"Neutrino Flavor Transformation in Neutron Star Mergers."* Phys.Rev.Lett. 135 (2025), 091401, (Editors' Suggestion, DOE Highlights, Penn State News) arXiv:2503.11758
4. **Yi Qiu***, Xisco Jiménez Forteza, Pierre Mourier, *"Linear vs. nonlinear modelling of black hole ringdowns."* Phys.Rev.D 109 (2024) 6, 064075, arXiv:2312.15904
5. **Yi Qiu**, Ke Wang, Jian-hua He*, *"Amplitude modulation in binary gravitational lensing of gravitational waves."* arXiv:2205.01682

PUBLICATIONS in prep (first-author only)

1. **Yi Qiu***, et al., *"Performance-portable moment based neutrino transport with AthenaK."* draft writing
2. **Yi Qiu***, et al., *"Long term GRMHD simulation of binary neutron stars merger with M1 neutrino transport."* in prep.
3. **Yi Qiu***, et al., *"Magnetic field amplification in high resolution GRMHD simulation of binary neutron stars merger with M1 neutrino transport"* in prep.

TALKS/SEMINARS

1. **Invited Seminar**, INFN-TIFPA seminar at TIFPA, University of Trento, Italy — Oct 19, 2026
2. **Talk**, ECT*-EMMI/GSI workshop at Trento, Italy — Sep 22, 2026
3. **Flash Talk**, AthenaK Summer School at Penn State University Campus — July 9, 2026
4. **Talk**, 25th Eastern Gravity Meeting at Rockefeller Hall, Cornell University — May 28, 2026
5. **Talk**, Collective Neutrino Oscillations in Supernovae and Neutron Star Mergers at Institute of Physics, Academia Sinica, Taiwan — Mar 26, 2026
6. **Talk**, APS Summit 2026 at Denver Convention Center — Mar 17, 2026
7. **Talk**, YITP long-term workshop: Multi-Messenger Astrophysics in the Dynamic Universe at Yukawa Institute for Theoretical Physics, Kyoto University — Feb 5, 2026
8. **Invited Seminar**, The Rising Researchers Seminar Series virtual at Institute for Nuclear Theory, University of Washington — Jan 20, 2026
9. **Virtual Talk**, Neutrino from Home — Dec 2, 2025

10. **Invited Seminar**, HEP/Astro Seminar at University of Tennessee, Knoxville, TN Nov 3, 2025
11. **Talk**, Gravitational-Wave Astronomy Meets Nuclear Astrophysics 2025 at California State University Fullerton, Fullerton, CA July 7, 2025
12. **Poster**, Caltech Relativistic Astrophysics Summer School 2025 at California Institute of Technology, Pasadena, CA June 23, 2025
13. **Workshop**, ALCF INCTIE GPU Hackthon at Argonne National Lab, Chicago, IL May 20, 2025
14. **Talk**, The 7th neighborhood workshop at Pennsylvania State University, PA April 3, 2025
15. **Talk**, APS Meeting 2025 at Anaheim, CA March 18, 2025
16. **Seminar**, Primordial Universe and Gravity (PUG) Seminar at Institute for Gravitation and the Cosmos, The Pennsylvania State University, PA Feb 21, 2025
17. **Invited Seminar**, University of Tennessee, Knoxville, TN Oct 29, 2024
18. **Poster**, CAMPS 2024 at University of Chicago, Chicago, IL August 2, 2024
19. **Talk**, The 6th neighborhood workshop at Pennsylvania State University, PA April 25, 2024
20. **Talk**, APS April Meeting 2024 at Sacramento, CA April 4, 2024
21. **Seminar**, Primordial Universe and Gravity (PUG) Seminar at Institute for Gravitation and the Cosmos, The Pennsylvania State University, PA March 15, 2024
22. **Talk**, N3AS Summer School at University of California, Santa Cruz, CA July 20, 2023
23. **Seminar**, Primordial Universe and Gravity (PUG) Seminar at Institute for Gravitation and the Cosmos, The Pennsylvania State University, PA May 5, 2023

AWARDS/ HONOURS/ SCHOLARSHIPS

China Scholarship Council Award for Outstanding Students Abroad ($6000) Jun. 2026
DOE-INFN Summer Exchange Fellowship (€6000) Apr. 2026
David. C. Duncan, Downsbrough, Troxell, and W. Donald Miller Graduate Fellowships Dec. 2025
Penn State Graduate Student International Travel Award ($1500) Nov. 2025
Edward M. Frymoyer Honors Award in the Eberly College of Science ($2100) Nov. 2025
Dean of the Graduate School Student Scholarship ($7000) Jun. 2025
Travel and hotel support for GravNu 2025 Workshop Presenter ($2000) Jun. 2025
Travel support for Caltech Relativistic Astrophysics Summer School ($800) Apr. 2025
DGRAV Student Travel Award ($300) Apr. 2024
David C. Duncan Graduate Fellowships ($2000) Mar. 2022
Highest merit scholarship (Top 1/41) Sep. 2020
University Undergraduate Physics Tournament (Top 1/32) Apr. 2019

SERVICES

American Physical Society
Member and referee of 2 manuscripts for Physical Review D 2022 – present

NSF Partnerships for Research and Education in Materials (PREM) program
Python/Machine Learning Instructor for African Americans in NCCU Sep 2023 – May 2024

Physics and Astronomy for Women+ (PAW+)
Professional Development Co-Chair Aug 2024 – Present

Physics Graduate Student Association (PGSA)
Web Master Jul 2024 – May 2025

Physics Graduate Peer Mentorship Program
Student Mentored: Harshraj Bandyopadhyay Sep 2024 – present

Research Experience for Undergraduates (REU) Program
Student Mentored: Bri Aleman, Davin Tipton Jun 2025 – Present